When does an LLM autoresearch loop find real signal?

A controlled study on twelve biomedical prediction tasks

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Abstract

Autoresearch loops, in which a large language model (LLM) repeatedly rewrites a single solution file and a change is kept only if a fixed development score improves, are an increasingly common way to build predictive pipelines. We ask when such a loop finds signal that holds up on held-out and external data. We ran a loop driven by Claude Opus 5.5 on 12 biomedical prediction tasks (scalp and neonatal EEG, resting-state fMRI, 16S microbiome, proteomics and drug response; 3 splits per task, 20 rounds per loop). Each champion was scored once on a held-out 25% of subjects and on eight external datasets, and compared with three baselines given the same 20 development evaluations: one LLM proposal, 20 independent LLM proposals without feedback, and Optuna search over a hand-written pipeline. Development scores improved in all 36 runs; held-out scores in 27. Averaged over 11 AUROC tasks the loop gained +0.052 AUROC over the seed (95% CI +0.009 to +0.095), compared with +0.002 for one proposal, +0.014 for 20 proposals without feedback and +0.035 for Optuna. Its advantage over Optuna (+0.017, $p = 0.29$) was significant only on abnormal-EEG and neonatal-seizure detection. When site and demographics were available, the loop built its models on them, and its gains on psychiatric EEG did not transfer to external cohorts. Feedback is what makes the LLM useful here, but the loop does not replace shortcut-free scoring and external validation.

1 Introduction

LLM agents can now write, run and revise machine-learning code with little supervision [10, 22, 23, 26]. The simplest version is the *autoresearch loop* [25, 32]: one editable file, one fixed metric, and a model that proposes a change each round, kept only if the metric improves. On development data the loop always looks like it works, because a keep-if-better rule over a fixed cross-validation (CV) score cannot make that score go down.

Biomedical prediction is a hard test of whether such gains are real. Datasets are small, cases and controls are often recruited and recorded differently, and site, scanner, batch and demographics can predict the label better than the biology [17, 37, 38]. Leakage and shortcuts are already common in human-written studies [24, 31], and selecting the best of many candidates on one CV score is itself a source of optimistic bias [9]. An optimizer that tries many ideas against one score has every opportunity to find all of these.

We therefore ask not whether the loop improves the development score, but *when* the improvement holds up: on held-out subjects, on external cohorts, and against simpler uses of the same budget. Across 12 tasks in six domains under one fixed protocol, we contribute (i) held-out results for 36 loop runs, each scored once; (ii) three same-budget baselines that separate the effect of feedback from that of the LLM and of ordinary search; (iii) multiplicity-corrected and pooled statistics with a sensitivity analysis; (iv) evidence that the loop exploits site and demographic shortcuts when

they are available; and (v) transfer to eight external datasets, run-to-run repeats and a transduction audit.

2 Related work

Benchmarks such as MLAgentBench [22] and MLE-bench [10] evaluate agents that write and run training code; AIDE [23] searches code with tree-structured refinement; the AI Scientist [26] automates the wider research cycle; and FunSearch [33] and AlphaEvolve [27] pair LLM proposals with an automatic evaluator. Most of this work reports scores on the benchmark the agent optimized. The autoresearch loop [25] is a deliberately minimal member of this family, and our protocol reuses that of radnyees [32]. As a non-LLM reference we use tree-structured Parzen estimators [7] in Optuna [2]. Our focus is a controlled evaluation in a domain where selection bias [9], leakage [24] and site or demographic shortcuts [8, 38] are well documented.

3 Methods

Loop. A fixed harness loads the data and defines labels, splits and metric; it is never shown to or edited by the LLM. A solution is one Python file implementing `featurize(items, ctx)`, which never receives labels, and `fit_predict(F_train, y_train, F_test, ctx)`, which receives training labels only. Each round the LLM (Claude Opus 5.5, reasoning effort *high*) receives a general brief, a project brief, the current best solution and the full history of previous rounds (rationale, declared feature/model families, score, kept or not, last failure), and returns a complete new solution. Candidates run on CPU containers with a 1,500 s limit (timeouts and crashes count as failed rounds). A loop runs 20 rounds with patience 10 and keeps a candidate iff its development score improves by more than 10^{-4}.

Tasks. Table 1 lists the 12 tasks: abnormal EEG (TUAB [28]); depression and ADHD (TDBRAIN [36]); schizophrenia (ASEEG [3]); per-second neonatal seizure detection (Helsinki [34]); autism and ADHD from fMRI connectivity (ABIDE I [15], ADHD-200; [29]); colorectal cancer (CRC) and IBD from 16S profiles (MicrobiomeHD [5, 16]); Alzheimer's disease (AD) from plasma [30] and CSF [4] proteomics; and drug response from CCLE expression to PRISM AUC [11, 13]. Seeds are deliberately standard: log band power with L2 logistic regression (scalp EEG); line length, RMS and band power with gradient boosting and smoothing (neonatal EEG); Fisher-z CC200 connectivity with logistic regression (fMRI); genus abundance with a random forest (16S); $\log_2$, median centring and logistic regression (proteomics); and ridge regression on the most variable genes (drug response).

Shortcut-free protocol (v2). Our first protocol (v1) passed age, sex, head motion and site to the solution and scored multi-site tasks with pooled AUROC; the loop used them (Section 4.4). All other results use v2: no demographics or motion; site, study and batch only for harmonization; multi-site tasks scored *within site* (size-weighted mean of per-site AUROCs); and TDBRAIN tasks restricted to the age range shared by patients and controls. v2 was designed after we saw the v1 results.

Evaluation. For each split seed (1, 2, 3), 25% of subjects (grouped by subject; stratified by label and site) form a held-out *Eval* set and the rest a *dev pool*. The loop optimizes *Dev*, a 5-fold grouped CV score inside the dev pool. After the loop, seed and champion are refit on the whole dev pool

Table 1: Tasks. Eval n = held-out subjects in split 1 (recordings for TUAB); other splits differ by at most a few. †Size-weighted mean of per-site (or per-study) AUROCs.

Task	Data	Label	Metric	Eval n (cases)	External sets
Abnormal EEG	TUAB v3.0.1, train split	abnormal vs. normal	AUROC	688 rec. / 519 pat. (346)	TUAB official eval
MDD	TDBRAIN, age-matched	MDD vs. healthy	AUROC	70 (38)	Mumtaz 2016
ADHD	TDBRAIN, adults	ADHD vs. healthy	AUROC	43 (13)	–
Schizophrenia	ASEEG	SZ vs. healthy	AUROC	25 (13)	MSU, RepOD
Neonatal seizure	Helsinki	seizure second (2/3 experts)	concat. AUROC	20 neonates	–
Autism	ABIDE I, 20 sites	ASD vs. TD	AUROC†	262 (129)	–
ADHD	ADHD-200, 8 sites	ADHD vs. TD	AUROC†	217 (82)	–
CRC	MicrobiomeHD, Baxter	CRC vs. healthy	AUROC	73 (30)	Zeller, Zackular, Xiang, Zhao
IBD	MicrobiomeHD, 4 studies	IBD vs. control	AUROC†	146 (117)	–
AD, plasma	PXD065835	AD vs. control	AUROC	22 (13)	–
AD, CSF	Bader 2020, 4 cohorts	clinical AD vs. non-AD	AUROC†	34 (12)	–
Drug response	CCLE → PRISM, 200 drugs	dose–response AUC	per-drug Pearson r	120 cell lines	–

and scored *once* on Eval and on external datasets. The three Eval sets overlap, so splits are not independent replicates.

Same-budget baselines. All arms start from the same seed, use the same harness, time limit and 20 Dev evaluations, keep a candidate only if it beats the seed on Dev, and are scored once on the same Eval sets. *One-shot*: one LLM proposal. *Best-of-20*: 20 independent LLM proposals that see the briefs and the seed but no history or scores, isolating the value of feedback. *Optuna*: 20 trials of multivariate TPE over a hand-written parameterized pipeline per project (which feature blocks to include, which classifier, its regularization). These search spaces were written *after* the main study and include feature families the loop had used, which makes this baseline conservative from the loop's point of view.

Statistics. Per task, the gain is the mean over three splits; its CI and two-sided p come from a paired bootstrap that resamples subjects from the union of the Eval sets (2,000 draws, $(b+1)/(B+1)$ per tail [12]), Holm-adjusted across 12 tasks [21]. This test treats the fitted models as fixed; it does not capture variation between splits or loop runs, which we report separately. Pooled over the 11 AUROC tasks, we report the equal-weight mean of task gains with a t-interval (primary) and DerSimonian–Laird random effects [14] (secondary). Run-to-run repeats, shortcut-only models and the transduction audit are described in Appendix A.

4 Results

4.1 Development gains are guaranteed; held-out gains are not

Across 36 runs the loop ran 710 rounds and kept 283 candidates (10 failed rounds). Dev improved in 36/36 runs, as the keep-if-better rule guarantees; Eval improved in 27/36, with per-split $p < 0.05$ in 15/36 (Table 4). The mean Dev gain (0.072) exceeded the mean Eval gain (0.049) and the two were only moderately correlated ($r = 0.38$; Figure 5). CRC, plasma AD and ABIDE autism lost or stayed flat on Eval despite consistent Dev gains (Table 2).

4.2 Feedback, not the LLM alone, produces the gains

Table 2 and Figure 1 compare the loop with the baselines. One LLM proposal was kept in 15/36 splits and changed Eval by +0.002 on average. Twenty proposals without feedback did better (+0.014) but were kept in only 25/36 splits: on neonatal seizure detection and ADHD (TDBRAIN),

Table 2: Held-out Eval per arm with the same budget of 20 Dev evaluations (bold: best non-seed arm), and the loop's gain over the seed and over Optuna with Holm-adjusted bootstrap p-values (bold gain: $p_{\text{Holm}} < 0.05$). splits ↑: splits where the loop beat the seed on Eval. †Within-site AUROC. ‡Mean per-drug Pearson r. §Gain carried by split 1, whose seed was below chance (split gains +0.264, +0.001, +0.015).

	Held-out Eval (mean of 3 splits)					Loop − seed		Loop − Optuna		
Task	Seed	One-shot	Best-of-20	Optuna	Loop	gain	p_{Holm}	gain	p_{Holm}	splits ↑
Abnormal EEG (TUAB)	0.847	0.866	0.888	0.889	**0.918**	**+0.071**	0.012	**+0.029**	0.012	3/3
MDD (TDBRAIN)	0.850	0.864	0.884	0.946	**0.954**	**+0.104**	0.012	+0.008	1.000	3/3
ADHD (TDBRAIN)	0.785	0.785	0.785	0.918	**0.922**	**+0.137**	0.012	+0.004	1.000	3/3
Schizophrenia (ASEEG)	0.677	0.677	0.705	0.678	**0.799**	+0.122	0.165	+0.121	0.504	3/3
Neonatal seizure (Helsinki)	0.827	0.827	0.827	0.824	**0.899**	**+0.072**	0.012	**+0.075**	0.012	3/3
Autism (ABIDE I)†	0.781	0.783	**0.790**	0.781	0.778	−0.002	0.915	−0.003	1.000	1/3
ADHD (ADHD-200)†§	0.572	0.570	0.627	0.650	**0.665**	**+0.093**	0.040	+0.015	1.000	3/3
CRC (Baxter)	0.795	**0.789**	0.787	0.759	0.740	**−0.055**	0.040	−0.020	1.000	0/3
IBD (4 studies)†	0.721	0.737	0.755	0.713	**0.755**	+0.035	0.312	+0.043	0.686	3/3
AD, plasma	0.912	0.895	0.897	**0.934**	0.866	−0.046	0.312	−0.068	0.220	0/3
AD, CSF†	0.913	0.913	0.891	**0.971**	0.954	+0.041	0.518	−0.017	1.000	2/3
Drug response (PRISM)‡	0.149	0.149	0.149	0.148	**0.160**	+0.012	0.162	+0.013	0.306	3/3

none of 120 independent proposals beat the seed on Dev, while the loop improved every split of both tasks. The loop's own first two rounds on neonatal seizures also failed in every split; improvement came once it could build on what had failed. All 720 best-of-20 proposals ran, so low adoption reflects proposal quality rather than a broken baseline.

Optuna gained +0.035 on average and matched the loop on both TDBRAIN tasks (MDD 0.946 vs. 0.954; ADHD 0.918 vs. 0.922) and on ADHD-200 (0.650 vs. 0.665); it beat the loop on plasma and CSF AD. The loop's advantage over Optuna was Holm-significant only on abnormal EEG (+0.029) and neonatal seizure detection (+0.075).

4.3 Corrected and pooled statistics

Against the seed, the loop's gain is Holm-significant on five tasks and its loss on CRC (Table 2). The ADHD-200 gain rests on one split (a t-test across splits gives $p = 0.39$), which the subject bootstrap does not capture. Pooled over 11 AUROC tasks (Table 3), the loop gained +0.052 over the seed ($p = 0.023$; $I^2 = 85\%$), +0.038 over best-of-20 ($p = 0.051$) and +0.017 over Optuna ($p = 0.29$). The gain over the seed depends on the TDBRAIN tasks, whose signal does not transfer (Section 4.5): without them it is +0.037 ($p = 0.11$), and +0.030 ($p = 0.21$) without ADHD-200 as well. Per-task results for all comparisons are in Table 5 and Figure 4.

4.4 The loop finds shortcuts when they exist

Under protocol v1, every ADHD-200, ABIDE and pooled-IBD champion referenced non-biological inputs: site, age and sex (ADHD-200); site, age, sex and head motion (ABIDE); study identity (IBD). These inputs carry substantial signal alone (Figure 2): site plus age and sex reach a pooled Eval AUROC of 0.775 on ADHD-200, close to the v1 champion (0.813); study identity alone gives 0.691 on IBD. Rescored within site, the v1 ADHD-200 champion drops from 0.813 to 0.731, close to the v2 champion (0.717). Within-site scoring removes the site shortcut but not the demographic one: on ADHD-200, age and sex alone give a within-site AUROC of 0.647, just below the v2

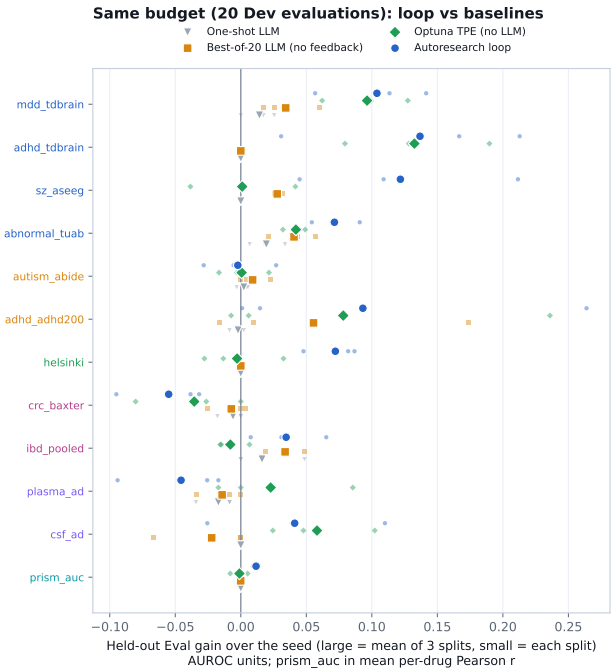


Figure 1: Held-out Eval gain over the seed per arm, same budget. Large markers: mean of 3 splits; small markers: individual splits.

Table 3: Pooled Eval gain over the 11 AUROC tasks: equal-weight mean with t-interval (primary) and DerSimonian–Laird random effects (secondary). Sensitivity rows drop the TDBRAIN tasks (separate recruitment; no transfer) and then ADHD-200 (gain carried by one split).

Comparison	k	mean gain	95% CI	p	random effects [95% CI]	I^2
Loop vs. seed	11	+0.052	[+0.009, +0.095]	0.023	+0.046 [+0.012, +0.080]	85%
Optuna vs. seed	11	+0.035	[+0.000, +0.070]	0.047	+0.027 [+0.001, +0.052]	70%
Best-of-20 vs. seed	11	+0.014	[−0.003, +0.031]	0.090	+0.012 [−0.003, +0.026]	74%
One-shot vs. seed	11	+0.002	[−0.005, +0.010]	0.457	+0.003 [−0.002, +0.008]	26%
Loop vs. Optuna	11	+0.017	[−0.017, +0.051]	0.290	+0.015 [−0.004, +0.034]	67%
Loop vs. best-of-20	11	+0.038	[−0.000, +0.076]	0.051	+0.028 [+0.003, +0.053]	83%
Loop vs. one-shot	11	+0.050	[+0.009, +0.090]	0.022	+0.041 [+0.011, +0.072]	85%
Sensitivity: leaving out tasks with suspect in-distribution gains						
Loop vs. seed, without TDBRAIN	9	+0.037	[−0.010, +0.084]	0.107		
Loop vs. seed, without TDBRAIN, ADHD-200	8	+0.030	[−0.021, +0.081]	0.210		
Loop vs. Optuna, without TDBRAIN	9	+0.019	[−0.024, +0.062]	0.328		
Loop vs. Optuna, without TDBRAIN, ADHD-200	8	+0.020	[−0.030, +0.070]	0.376		

champion's 0.665. Removing demographic *inputs* does not stop a model from using demographic *signal* encoded in the imaging data.

4.5 In-distribution gains did not transfer to other cohorts

Applied unchanged to external data (Figure 3, Table 6), only abnormal EEG transferred: on the official TUAB evaluation split the champion beat the seed in all three splits (AUROC 0.867 → 0.928, all CIs excluding zero). That split is from the same hospital, so it is a near-in-distribution check. The TDBRAIN depression seed was *inverted* on the Mumtaz cohort (AUROC 0.25–0.29) and the champions reached only 0.28–0.62. The numerical gain moves an inverted model towards

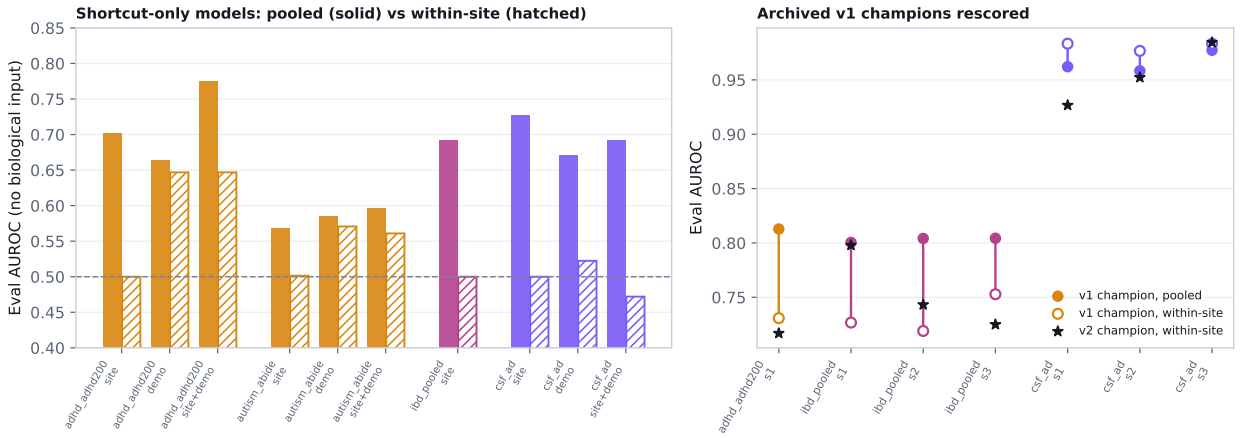


Figure 2: Left: Eval AUROC of shortcut-only models (no biological input), pooled (solid) and within site (hatched). Right: archived v1 champions rescored pooled and within site, with the v2 champion.

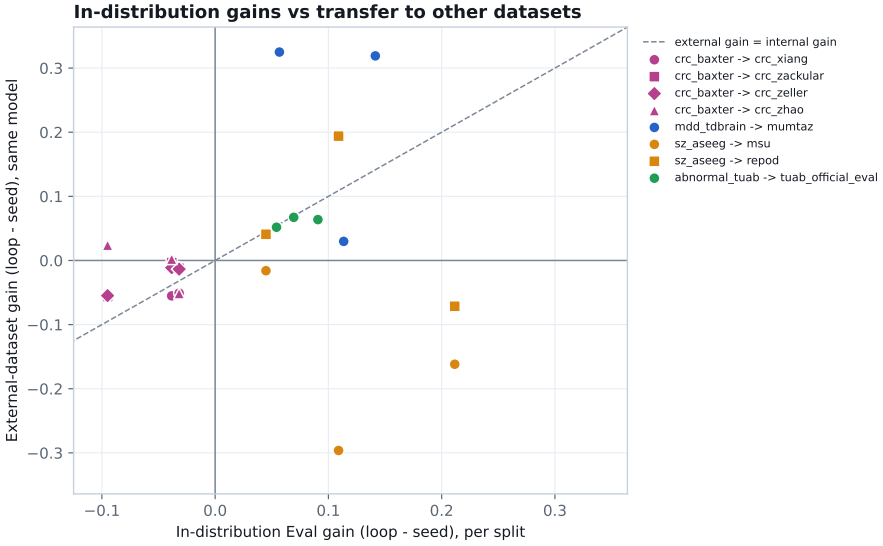


Figure 3: In-distribution Eval gain vs. external-dataset gain for the same model, one point per split and external set. Dashed: equal gain.

chance, which points to dataset-specific case/control signal rather than a transferable marker. The schizophrenia champion was worse than the seed on the MSU cohort in all three splits (0.696 → 0.538) and mixed on RepOD; the CRC champion lost on all four external cohorts on average.

4.6 Variance and transduction

Repeating the loop on the same split varied the Eval gain less than drawing a new split (run-to-run SD median 0.016 vs. split-to-split 0.042, smaller for 6 of 7 tasks; Figure 6). The sign was stable on IBD, MDD and schizophrenia and flipped between repeats on CRC, CSF AD and drug response; repeats were not affordable for abnormal EEG or neonatal seizure. All seeds were exactly inductive; scoring Eval inside a reference batch built from the dev pool changed champion scores by +0.001

on average (worst −0.024), so reported scores do not exploit Eval-set statistics (Figure 7). Eleven champions output batch-relative ranks and would need a reference cohort to score single patients.

5 Discussion

When the loop helped. The clearest gains came on abnormal EEG and neonatal seizure detection: signal-processing tasks with large or well-annotated data and clinically meaningful labels. They were the only tasks where the loop beat a loop-informed Optuna search, and on both, independent LLM proposals almost never beat the seed. The active ingredient is feedback, not the LLM's prior knowledge alone.

When it did not. Elsewhere the loop was not reliably better than ordinary search, and on three tasks it was worse than the seed on held-out data despite consistent Dev gains. Its largest in-distribution gains, on TDBRAIN, were matched by Optuna and did not transfer, pointing to dataset-specific signal that any optimizer would find. Given site and demographics, the loop used them. A held-out set scored once, a non-LLM baseline, shortcut-only models and an external cohort each changed our reading of at least one task; the Dev curve alone would have changed none. Our numbers are also not better than published results where a rough comparison is possible (Helsinki 0.899 vs. about 0.955 [35]; on CRC the seed already matched the MicrobiomeHD baseline [16]).

Limitations. Several Eval sets are small (20–43 subjects), the three splits overlap, and repeats cover only 7 of 12 tasks. We used one LLM at one setting and a CPU budget that favours light models. The Optuna spaces were written with knowledge of the loop's results. TDBRAIN cases and controls were recorded separately, and ADHD-200 groups differ in sex and age within sites. Protocol v2 was designed after seeing v1. Further dataset caveats are in Appendix B.

6 Conclusion

An LLM autoresearch loop with feedback improved held-out performance over a simple seed on most of 12 biomedical tasks, clearly more than the same LLM without feedback, but significantly more than a tuned non-LLM search on only two signal-processing tasks. The same loop exploited site and demographic shortcuts when allowed, and its psychiatric-EEG gains did not transfer. Gains from such a loop are a hypothesis to test on held-out, shortcut-free and external data, not evidence of a better biomarker.

Data and code availability

Code, all loop records, champions, Eval predictions and robustness analyses: `https://github.com/radnyees/justautoeverything`. Raw data are not redistributed (TDBRAIN data use agreement, TUH/NEDC terms, CC BY-NC licences); `projects/*/data/` contains download and preprocessing scripts. Compute: $117.65 on Modal CPU containers (LLM calls not included).

A Additional methods

Prompt and rules. The brief asks for one well-motivated change per round, forbids reading anything outside what the harness passes in, requires inner CV (grouped by subject) for any tuning and lists the available libraries (numpy, scipy, scikit-learn, LightGBM, XGBoost, CatBoost, MNE, pyRiemann, nilearn, PyTorch on CPU, among others). Candidates run on Modal containers with 2–16 cores depending on the task.

Run-to-run variance. For the seven cheapest tasks, split 1 was rerun twice more from scratch (same seed, Dev CV and Eval set; only the LLM's sampling differs).

Shortcut-only models. On the four multi-site tasks, L2 logistic regressions using only site or study one-hot codes and/or demographics were fit on the dev pool and scored on Eval with both the pooled (v1) and within-site (v2) metric.

Transduction audit. The Eval protocol featurizes dev-pool and Eval items in one call and passes the whole Eval batch to `fit_predict`. For every seed and champion we tested (i) whether dev-pool features change when Eval items are featurized with them; (ii) a *chunked* score, with Eval split into 20 chunks (10 for large tasks), each featurized with the dev pool and predicted on its own; and (iii) a *reference-batch* score, as (ii) but each chunk predicted inside a batch of dev pool plus chunk, so batch statistics come from training-time data.

Pooled statistics. Drug response (Pearson r) is excluded from pooling. For random effects, task standard errors are floored at 0.005. Benjamini–Hochberg adjusted values [6] are in the released files.

B Dataset caveats

- **TDBRAIN.** All controls are adults, so tasks are age-matched to the shared range (ADHD: adults only); age is never given to the model. Patients and controls were recruited and recorded separately, so state or protocol differences cannot be excluded.
- **ASEEG.** The channel order is undocumented and was inferred from data (its correlation structure matches labelled RepOD recordings: $r = 0.855$ for the standard 10–20 order, $r < 0.1$ for the alternatives). MSU is adolescents with 1-min, 16-channel recordings.
- **TUAB.** First 60 s skipped, next 5 min used, 19 channels at 100 Hz, clipped at $\pm 800\,\mu$V [18].
- **Helsinki.** Label = majority of three experts per second; published results often use leave-one-out or other annotation rules [20, 35].
- **ABIDE / ADHD-200.** Within-site AUROC is stricter than the pooled metrics usually reported [1, 19].
- **Proteomics.** Plasma: single centre, the only open labelled AD plasma MS set we found. CSF label = clinical diagnosis.
- **Drug response.** 200 most variable drugs ($\geq$70% coverage) from the PRISM secondary screen; per-drug Pearson r on held-out cell lines.

C Additional results

Table 4: Seed vs. loop champion per task (means over 3 splits), Dev and Eval. "improved": splits with Eval gain > 0; "sig.": splits with one-sided per-split bootstrap $p < 0.05$. Symbols as in Table 2.

		Dev (5-fold CV)		Held-out Eval			
Modality	Task	seed	champion	seed	champion	Eval gain (mean ± sd)	improved / sig.
EEG	Abnormal EEG (TUAB)	0.865	0.934	0.847	0.918	+0.071 ± 0.018	3/3 / 3/3
	MDD (TDBRAIN)	0.827	0.927	0.850	0.954	+0.104 ± 0.043	3/3 / 3/3
	ADHD (TDBRAIN)	0.731	0.882	0.785	0.922	+0.137 ± 0.095	3/3 / 2/3
	Schizophrenia (ASEEG)	0.806	0.883	0.677	0.799	+0.122 ± 0.084	3/3 / 1/3
	Neonatal seizure (Helsinki)	0.744	0.858	0.827	0.899	+0.072 ± 0.021	3/3 / 3/3
fMRI	Autism (ABIDE I)†	0.740	0.787	0.781	0.778	−0.002 ± 0.028	1/3 / 0/3
	ADHD (ADHD-200)†§	0.591	0.666	0.572	0.665	+0.093 ± 0.148	3/3 / 1/3
16S	CRC (Baxter)	0.755	0.789	0.795	0.740	−0.055 ± 0.035	0/3 / 0/3
	IBD (4 studies)†	0.742	0.786	0.721	0.755	+0.035 ± 0.029	3/3 / 1/3
Proteomics	AD, plasma	0.884	0.939	0.912	0.866	−0.046 ± 0.042	0/3 / 0/3
	AD, CSF†	0.859	0.943	0.913	0.954	+0.041 ± 0.068	2/3 / 1/3
Expression	Drug response (PRISM)‡	0.138	0.156	0.149	0.160	+0.012 ± 0.002	3/3 / 0/3

Table 5: Per-task Eval differences (mean over 3 splits) with Holm-adjusted two-sided bootstrap p-values across 12 tasks; bold: $p_{\text{Holm}} < 0.05$.

	Loop − seed		Optuna − seed		Loop − best-of-20		Loop − Optuna	
Task	gain	p_{Holm}	gain	p_{Holm}	gain	p_{Holm}	gain	p_{Holm}
Abnormal EEG (TUAB)	**+0.071**	0.012	**+0.042**	0.012	**+0.031**	0.012	**+0.029**	0.012
MDD (TDBRAIN)	**+0.104**	0.012	**+0.096**	0.012	**+0.070**	0.048	+0.008	1.000
ADHD (TDBRAIN)	**+0.137**	0.012	+0.132	0.090	**+0.137**	0.012	+0.004	1.000
Schizophrenia (ASEEG)	+0.122	0.165	+0.001	1.000	+0.094	0.182	+0.121	0.504
Neonatal seizure (Helsinki)	**+0.072**	0.012	−0.003	1.000	**+0.072**	0.012	**+0.075**	0.012
Autism (ABIDE I)†	−0.002	0.915	+0.001	1.000	−0.011	0.572	−0.003	1.000
ADHD (ADHD-200)†§	**+0.093**	0.040	+0.078	0.351	+0.038	0.222	+0.015	1.000
CRC (Baxter)	**−0.055**	0.040	−0.036	0.568	**−0.048**	0.045	−0.020	1.000
IBD (4 studies)†	+0.035	0.312	−0.008	1.000	+0.001	0.957	+0.043	0.686
AD, plasma	−0.046	0.312	+0.023	1.000	−0.031	0.344	−0.068	0.220
AD, CSF†	+0.041	0.518	+0.058	1.000	+0.063	0.344	−0.017	1.000
Drug response (PRISM)‡	+0.012	0.162	−0.001	1.000	+0.012	0.260	+0.013	0.306

What the loop built. Figure 11 shows the feature and model families in the final champions as *declared by the LLM*; it is a self-report, not an analysis of the code. Champions typically extended the seed's representation with domain features: finer clinical sub-bands, aperiodic slope, connectivity, complexity and Riemannian covariance features (EEG); per-neonate and per-channel background normalization and multi-scale temporal context (neonatal seizures); log-ratio transforms and batch correction (microbiome); known AD proteins and pathway aggregates (proteomics). Ensembles of a linear model with boosting or kernel methods were common.

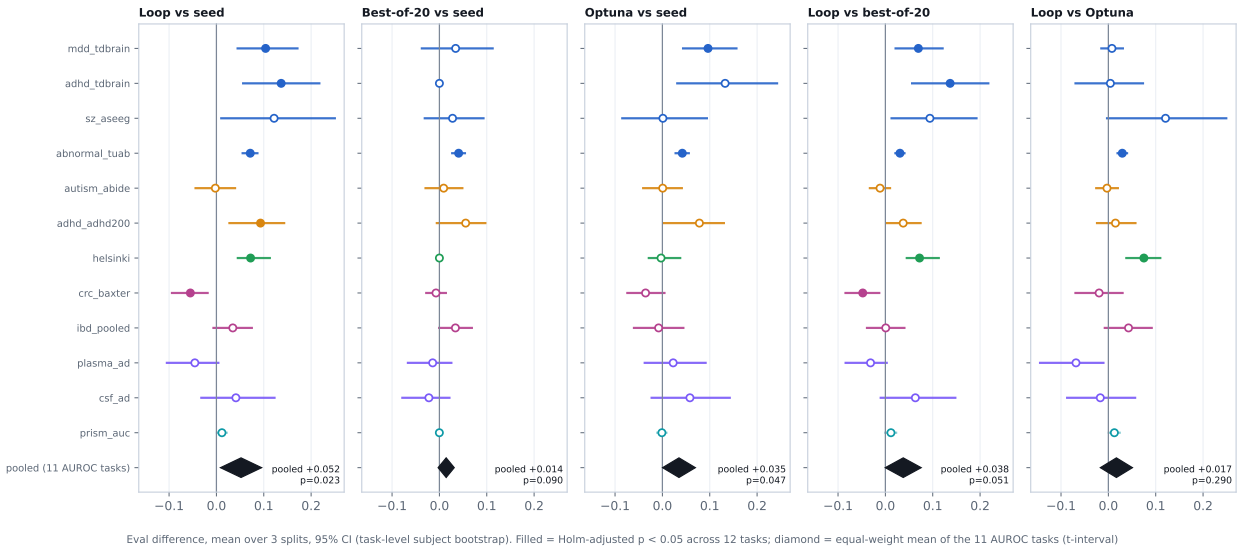


Figure 4: Per-task Eval differences with 95% CIs (task-level subject bootstrap) for five comparisons. Filled: Holm-adjusted $p < 0.05$; diamond: equal-weight mean of the 11 AUROC tasks.

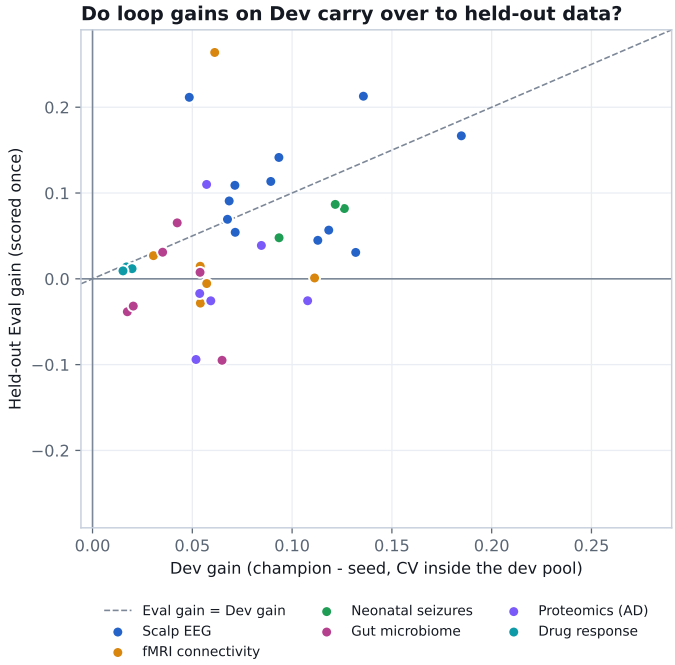


Figure 5: Dev gain vs. held-out Eval gain, one point per task and split. Dashed: Eval gain = Dev gain.



Figure 6: Eval gain of three independent loops on split 1 (filled) and of the loops on splits 2–3 (hollow), for the seven tasks with repeats.

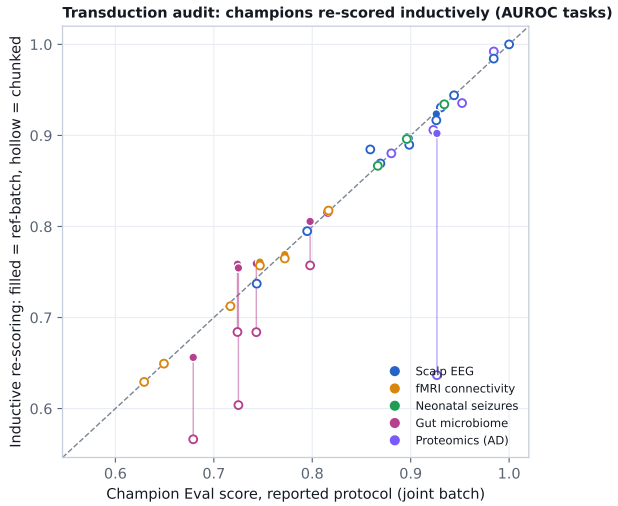


Figure 7: Transduction audit of the champions (AUROC tasks). x: reported Eval score (joint batch). Filled: reference-batch score; hollow: chunks scored in isolation.

Table 6: External datasets: seed and champion applied unchanged; 95% CI from the paired bootstrap.

Task	External set	split	seed	champion	gain	95% CI
Abnormal EEG (TUAB)	TUAB official eval	1	0.867	0.930	+0.064	[+0.034, +0.096]
		2	0.865	0.933	+0.067	[+0.035, +0.104]
		3	0.869	0.921	+0.052	[+0.020, +0.084]
MDD (TDBRAIN)	Mumtaz 2016	1	0.250	0.280	+0.030	[−0.123, +0.190]
		2	0.268	0.587	+0.319	[+0.120, +0.518]
		3	0.290	0.615	+0.325	[+0.154, +0.502]
Schizophrenia (ASEEG)	MSU	1	0.696	0.680	−0.016	[−0.108, +0.077]
		2	0.681	0.520	−0.162	[−0.276, −0.043]
		3	0.710	0.414	−0.296	[−0.438, −0.147]
	RepOD	1	0.490	0.531	+0.041	[−0.182, +0.262]
		2	0.500	0.429	−0.071	[−0.231, +0.109]
		3	0.505	0.699	+0.194	[−0.052, +0.448]
CRC (Baxter)	Zeller	1	0.813	0.801	−0.011	[−0.058, +0.035]
		2	0.818	0.805	−0.013	[−0.054, +0.024]
		3	0.846	0.791	−0.055	[−0.114, −0.002]
	Zackular	1	0.792	0.789	−0.003	[−0.053, +0.053]
		2	0.802	0.790	−0.012	[−0.055, +0.034]
		3	0.796	0.739	−0.056	[−0.144, +0.024]
	Xiang	1	0.748	0.693	−0.055	[−0.126, −0.007]
		2	0.763	0.712	−0.051	[−0.163, +0.054]
		3	0.738	0.685	−0.053	[−0.133, +0.012]
	Zhao	1	0.664	0.666	+0.002	[−0.029, +0.031]
		2	0.742	0.691	−0.051	[−0.112, +0.006]
		3	0.702	0.726	+0.024	[−0.043, +0.091]

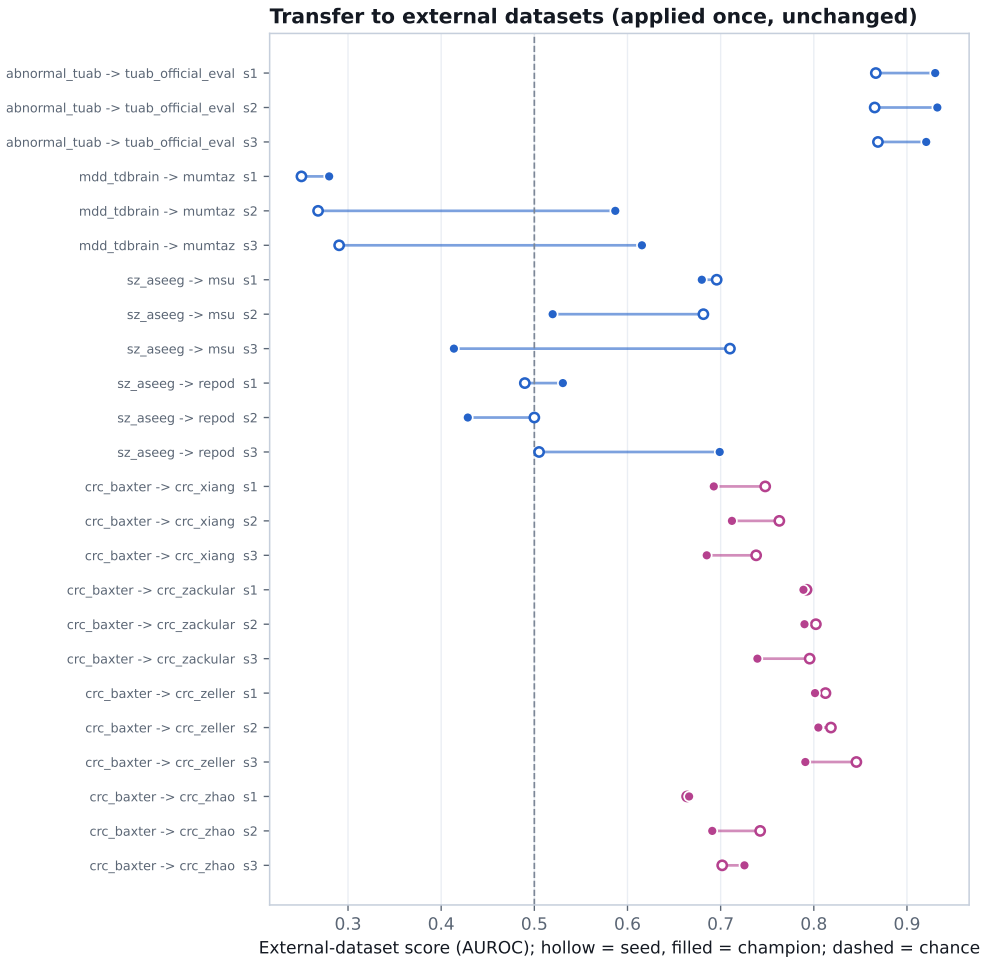


Figure 8: External-dataset scores. Hollow: seed; filled: champion; dashed: chance.

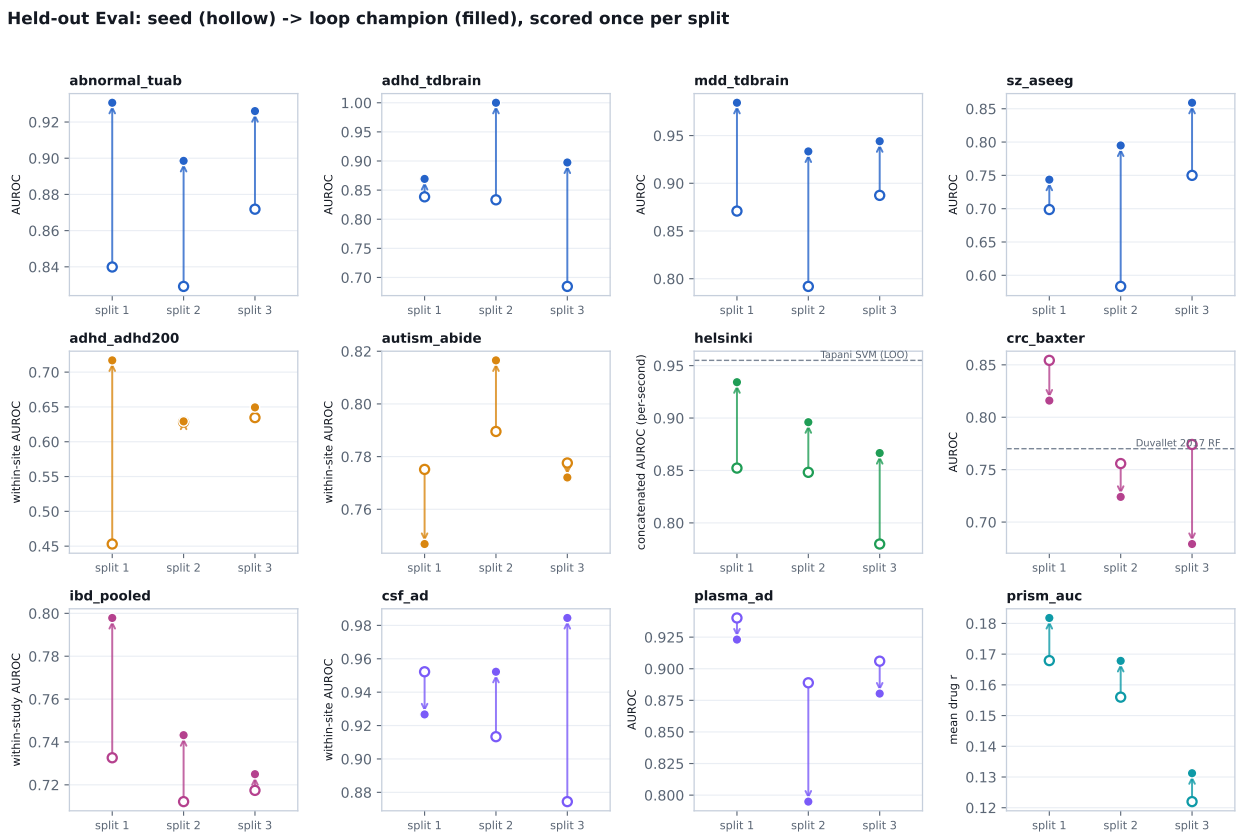


Figure 9: Held-out Eval score of the seed (hollow) and the loop champion (filled), per split. Dashed: published results on the same dataset (different protocols).

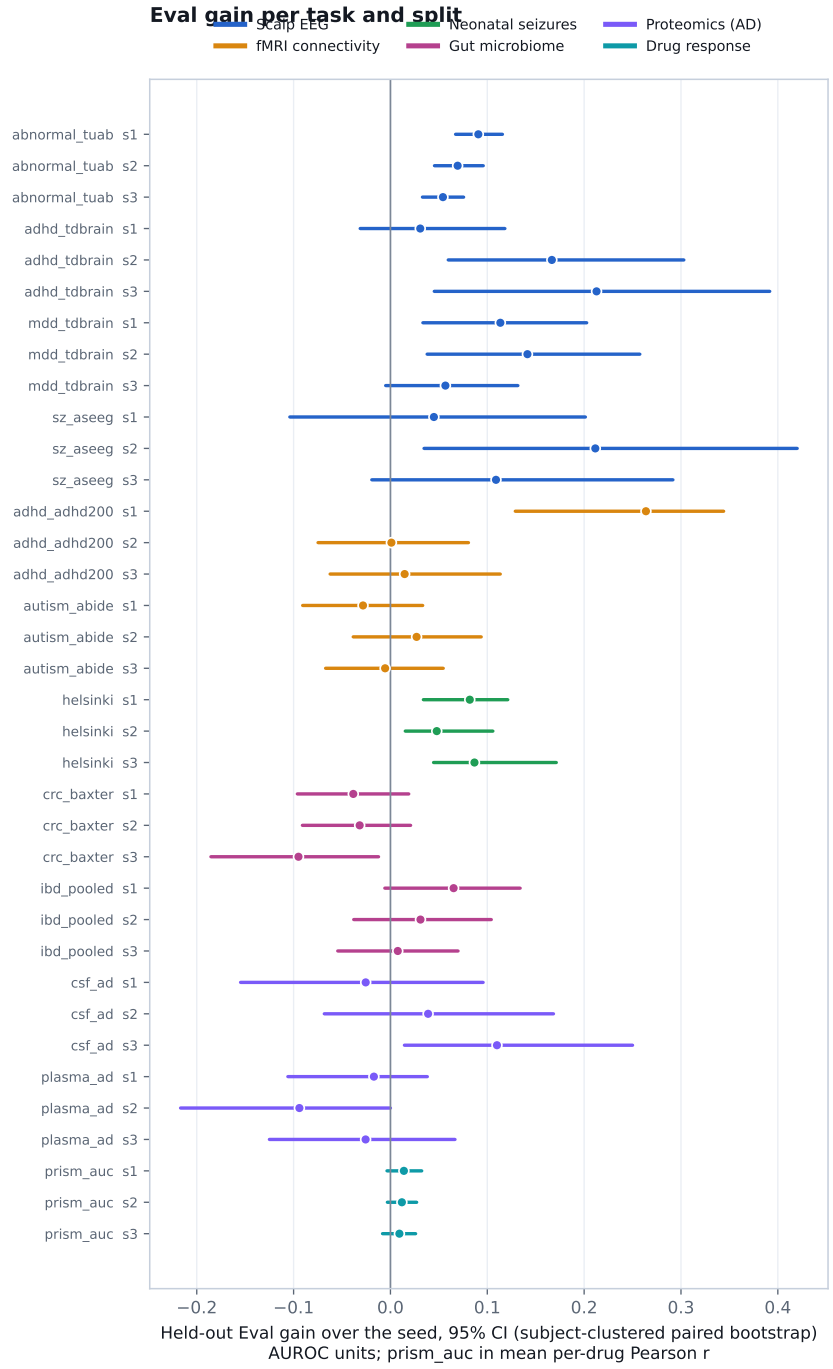


Figure 10: Held-out Eval gain over the seed with 95% CI (per-split paired bootstrap), per task and split.

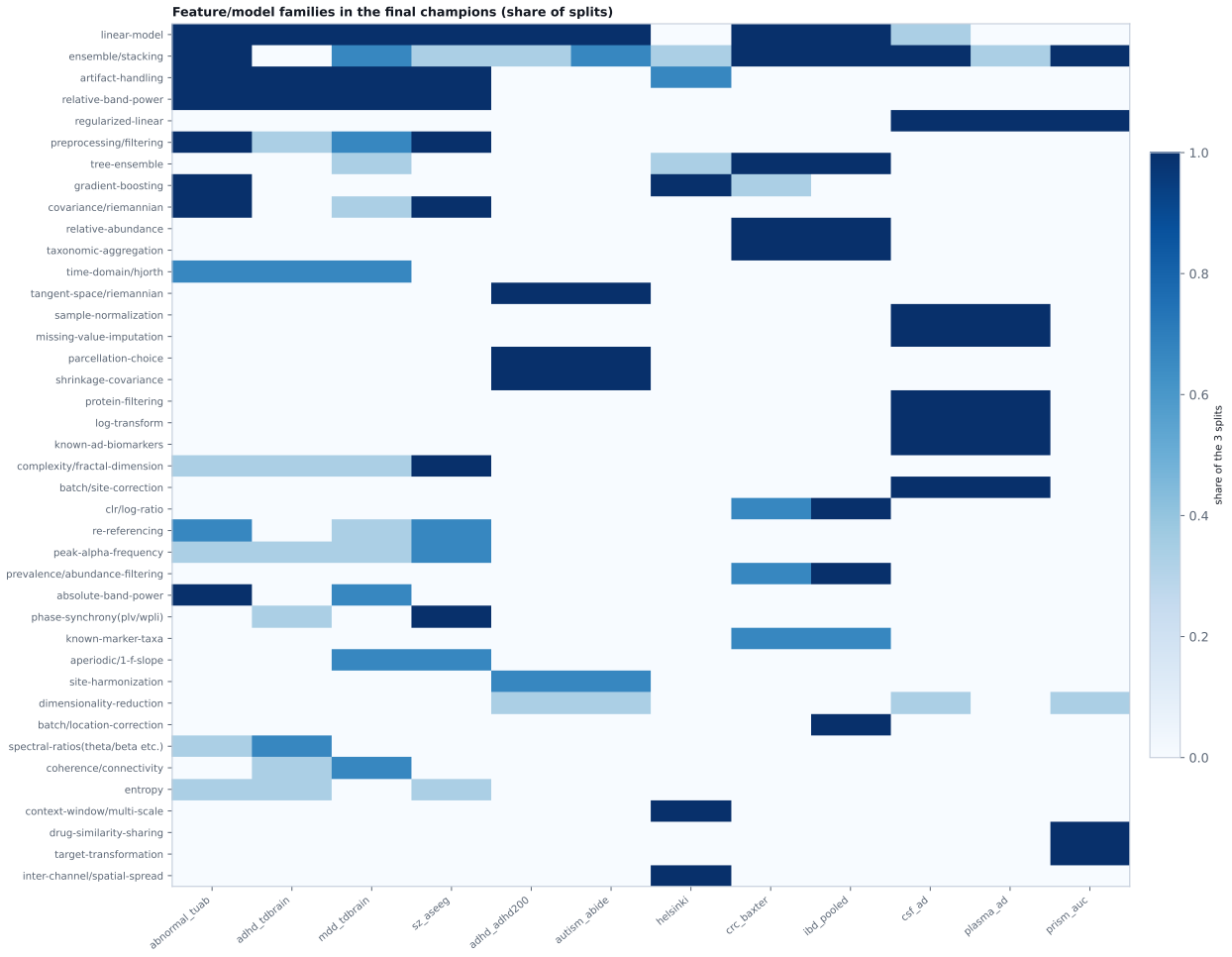


Figure 11: Feature/model families in the final champions (share of the 3 splits), as declared by the LLM.

Table 7: Every run. "failed" = timeouts + crashes. Eval gain CI and one-sided p from the per-split paired bootstrap (2,000 resamples).

Task	split	rounds	kept	failed	Dev gain	Eval gain	95% CI	p (one-sided)
Abnormal EEG (TUAB)	1	20	11	0	+0.068	+0.091	[+0.067, +0.116]	<0.001
	2	20	9	1	+0.068	+0.069	[+0.045, +0.096]	<0.001
	3	20	10	3	+0.071	+0.054	[+0.033, +0.076]	<0.001
MDD (TDBRAIN)	1	20	7	0	+0.089	+0.113	[+0.033, +0.203]	<0.001
	2	20	11	0	+0.093	+0.141	[+0.038, +0.258]	0.001
	3	20	6	0	+0.118	+0.057	[−0.005, +0.132]	0.042
ADHD (TDBRAIN)	1	20	6	0	+0.132	+0.031	[−0.031, +0.118]	0.186
	2	20	10	0	+0.185	+0.167	[+0.060, +0.303]	<0.001
	3	20	11	0	+0.136	+0.213	[+0.045, +0.392]	0.004
Schizophrenia (ASEEG)	1	20	6	0	+0.113	+0.045	[−0.104, +0.201]	0.307
	2	20	10	0	+0.048	+0.212	[+0.035, +0.420]	0.012
	3	20	8	0	+0.071	+0.109	[−0.019, +0.292]	0.065
Neonatal seizure (Helsinki)	1	20	11	0	+0.126	+0.082	[+0.034, +0.121]	0.002
	2	20	10	0	+0.093	+0.048	[+0.015, +0.106]	0.003
	3	20	7	1	+0.122	+0.087	[+0.045, +0.171]	<0.001
Autism (ABIDE I)[†]	1	19	7	1	+0.054	−0.028	[−0.091, +0.033]	0.816
	2	11	1	0	+0.031	+0.027	[−0.038, +0.094]	0.210
	3	20	10	0	+0.057	−0.005	[−0.067, +0.055]	0.564
ADHD (ADHD-200)[†§]	1	20	5	0	+0.061	+0.264	[+0.129, +0.344]	<0.001
	2	20	5	0	+0.111	+0.001	[−0.075, +0.081]	0.491
	3	20	7	0	+0.054	+0.015	[−0.062, +0.113]	0.306
CRC (Baxter)	1	20	5	0	+0.017	−0.038	[−0.096, +0.019]	0.922
	2	20	6	0	+0.020	−0.032	[−0.091, +0.021]	0.873
	3	20	10	0	+0.065	−0.095	[−0.185, −0.012]	0.991
IBD (4 studies)[†]	1	20	7	1	+0.042	+0.065	[−0.006, +0.134]	0.035
	2	20	6	1	+0.035	+0.031	[−0.038, +0.104]	0.185
	3	20	9	1	+0.054	+0.008	[−0.054, +0.070]	0.406
AD, plasma	1	20	6	0	+0.054	−0.017	[−0.106, +0.038]	0.790
	2	20	8	0	+0.052	−0.094	[−0.217, +0.000]	0.999
	3	20	6	0	+0.059	−0.026	[−0.125, +0.067]	0.772
AD, CSF[†]	1	20	10	0	+0.108	−0.026	[−0.155, +0.096]	0.692
	2	20	6	1	+0.085	+0.039	[−0.068, +0.168]	0.257
	3	20	8	0	+0.057	+0.110	[+0.014, +0.250]	0.015
Drug response (PRISM)[‡]	1	20	9	0	+0.017	+0.014	[−0.004, +0.032]	0.055
	2	20	9	0	+0.020	+0.012	[−0.003, +0.027]	0.059
	3	20	10	0	+0.015	+0.009	[−0.008, +0.026]	0.174

D Loop trajectories

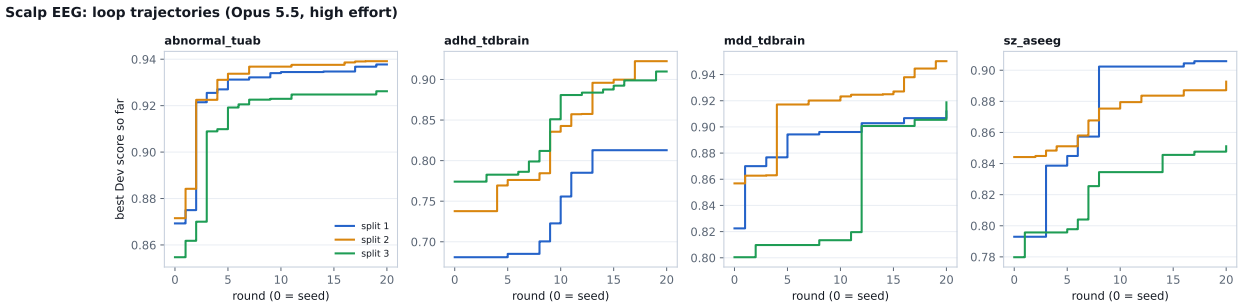


Figure 12: Best Dev score so far per round, scalp EEG tasks.

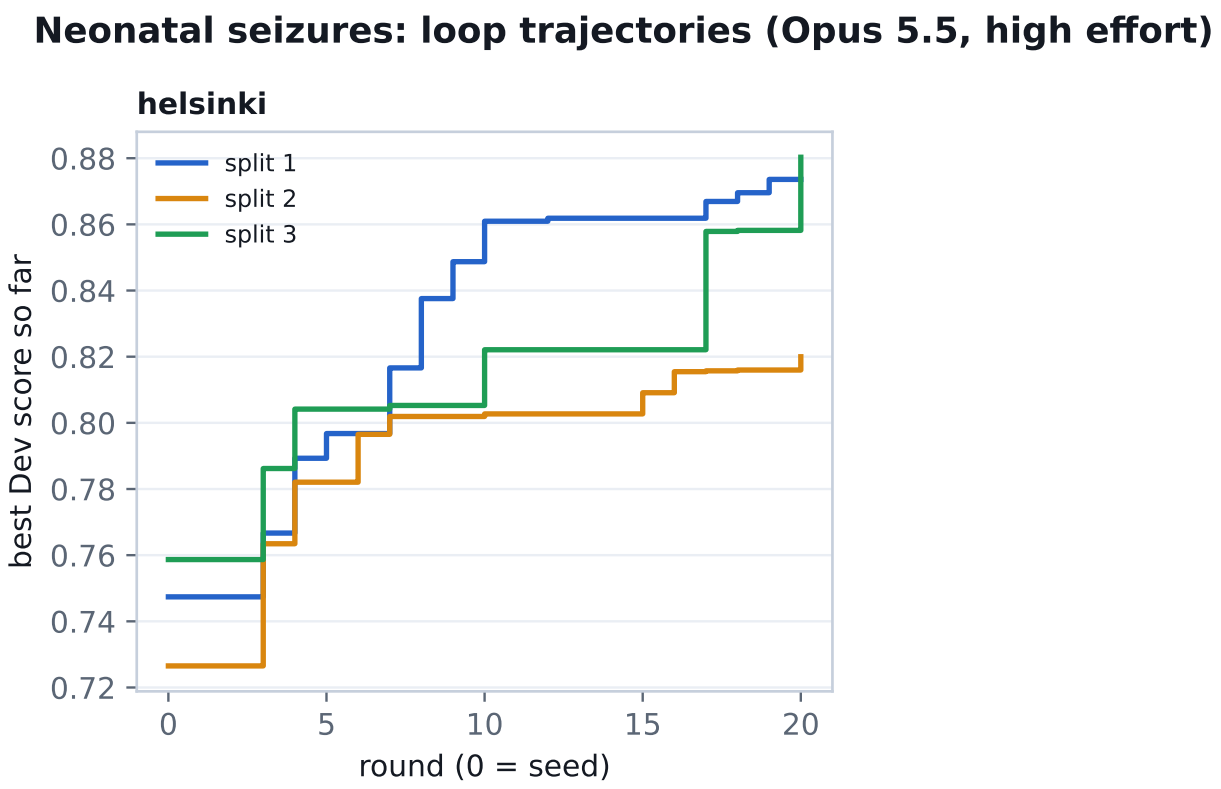


Figure 13: Best Dev score so far per round, neonatal seizure detection.

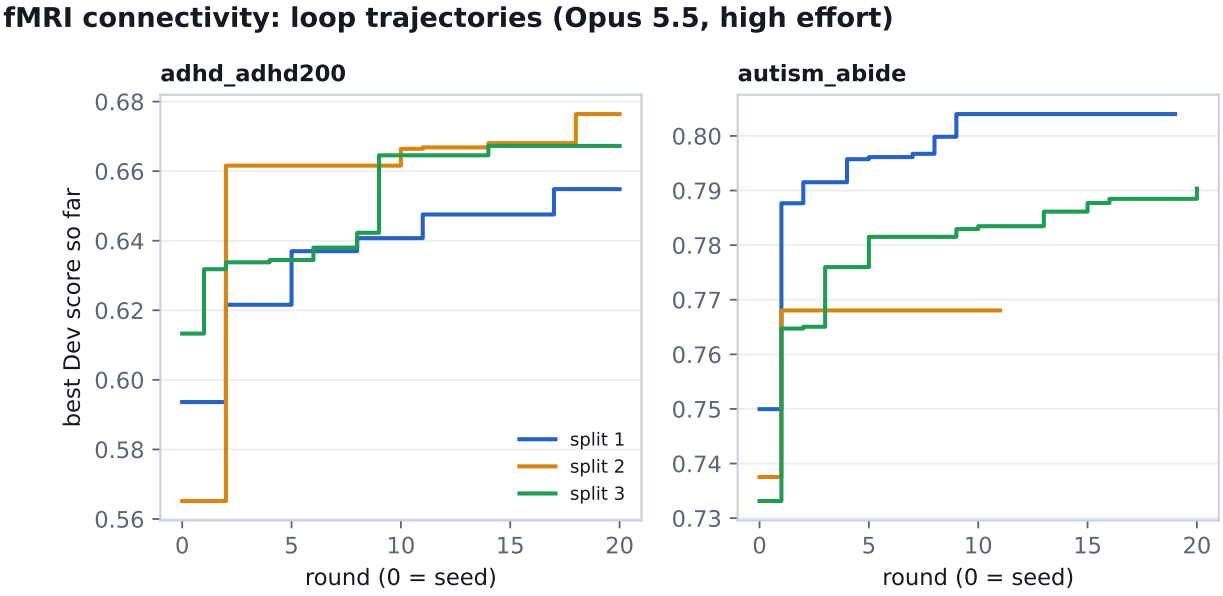


Figure 14: Best Dev score so far per round, fMRI connectivity tasks.

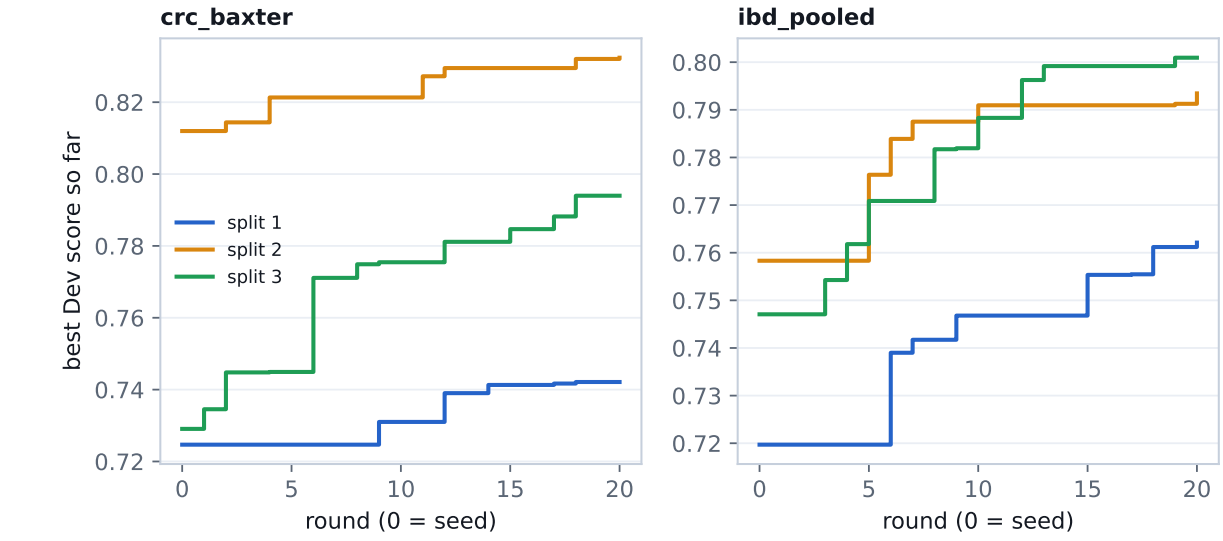


Figure 15: Best Dev score so far per round, gut microbiome tasks.

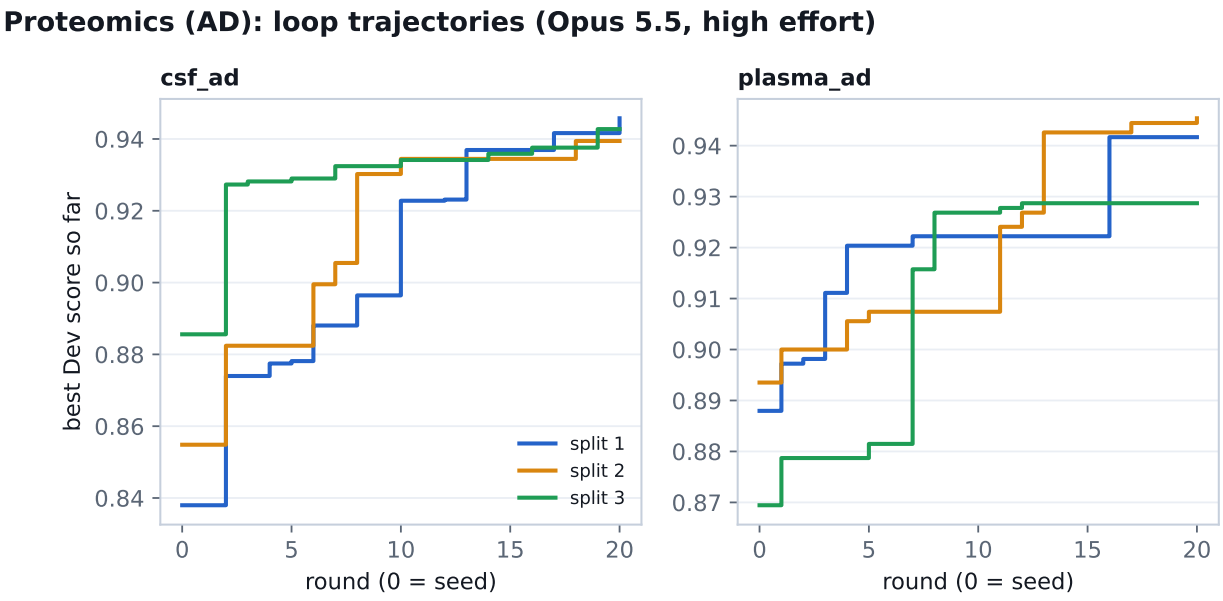


Figure 16: Best Dev score so far per round, proteomics (AD) tasks.

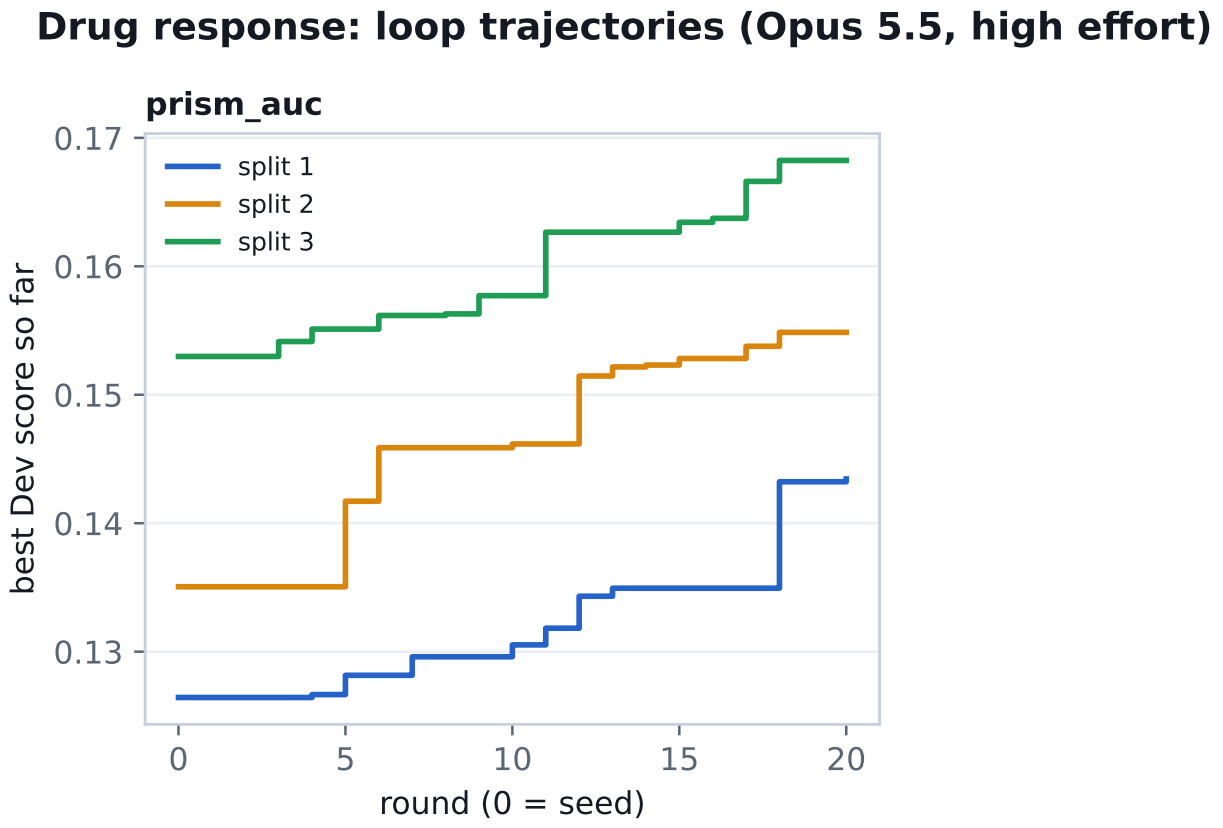


Figure 17: Best Dev score so far per round, drug response.

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